

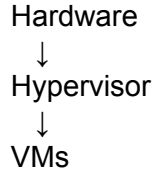
what is hypervisor?

A hypervisor is software that allows one physical computer to run multiple virtual machines (VMs) and manages the physical hardware resources given to each VM.

Two main types

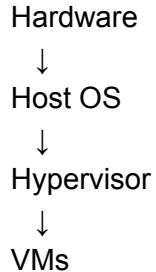
Type 1 — Bare Metal

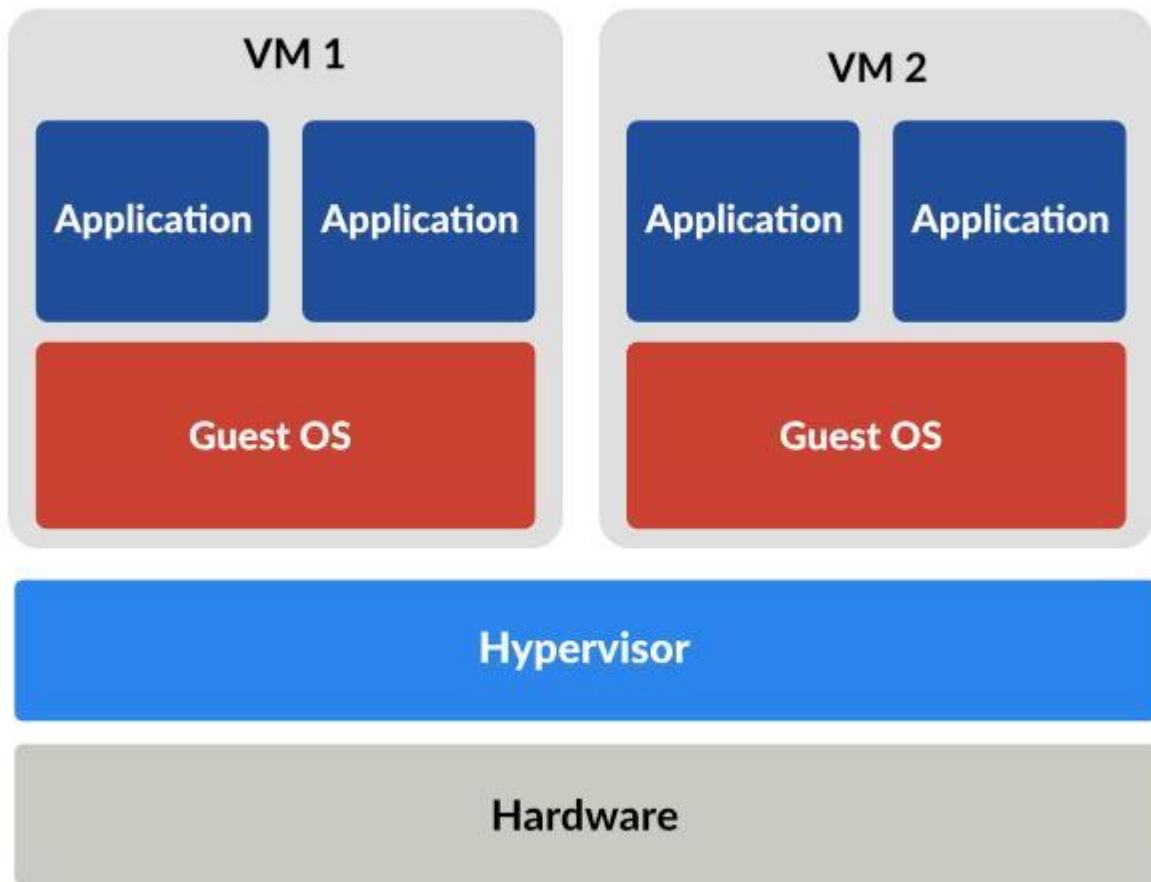
The hypervisor operates directly on the hardware:



Type 2 — Hosted

The hypervisor runs on top of a normal operating system:





Type 1

Examples

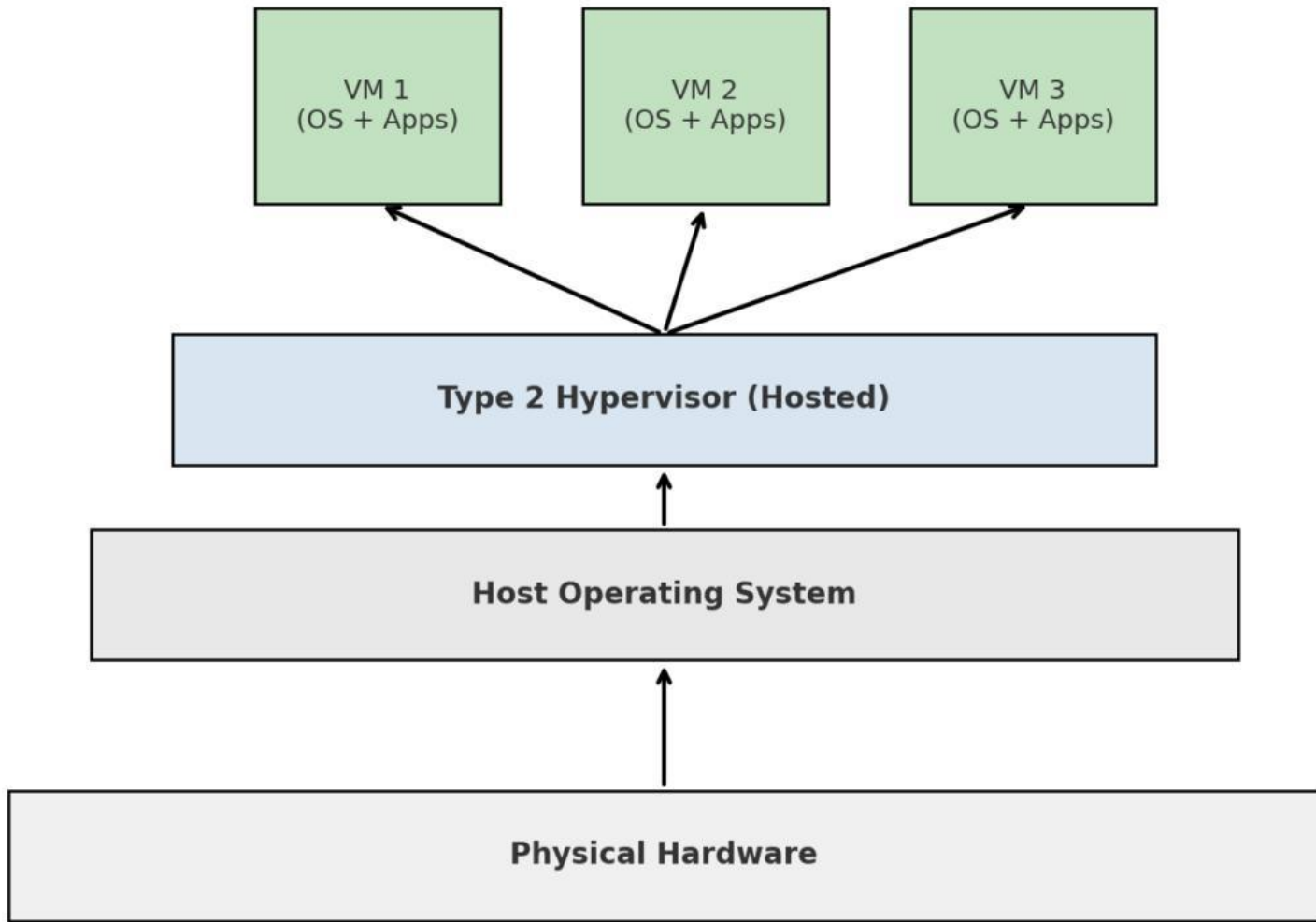
- VMware ESXi
- Microsoft Hyper-V
- Xen
- KVM*

Why is it called "bare metal"?

Because the hypervisor has direct access to the physical hardware.

This generally provides:

- High performance
- Low overhead
- Better resource management
- Better isolation
- Commonly used in **data centers and cloud infrastructure**



2. Type 2 Hypervisor — "Hosted"

Examples

- Oracle VirtualBox
- VMware Workstation
- VMware Fusion
- Parallels Desktop

KVM turns the **Linux kernel into a hypervisor** using hardware virtualization technologies such as:

- Intel VT-x
- AMD-V

KVM itself provides the core virtualization infrastructure, while tools such as **QEMU** provide device emulation and VM management.

KVM

CPU/hardware virtualization

QEMU

Provides virtual machine hardware and runs the VM

libvirt

Management layer

virsh

CLI client for libvirt

virt-manager

GUI client for libvirt

bridge-utils

Older/common bridge-management utilities; not required for basic NAT setup

